



M.Sc. Thesis

LITERATURE INSIGHTS SLIDE

Supervisor: Dr Deepak Ajwani

Student: Ashwin Prasanth

Week- 1,2

PAPERS READ:

Paper / Source	Core Idea / Contribution	Key Techniques	Relevance to Dissertation
LwD (Ahn et al., 2020)	Learning what to defer: The agent prioritizes "easy" decisions and defers hard ones to later stages.	Deep Reinforcement Learning (DRL); Diversification bonuses for solution variety.	Validates the strategy of prioritizing certain cell assignments to simplify the remaining crossword grid.
ConRep4CO (Guo et al.)	Uses SAT as a unified intermediary to learn representations across different problem types through contrastive learning.	Contrastive Representation Learning; augmentation-free alignment between original form and SAT.	Proves that complex constraints can be mapped to a universal representation (SAT), which crosswords can leverage.
Khalil et al. (2016)	The first successful framework to learn to branch on-the-fly by imitating Strong Branching (SB).	Learning-to-Rank; SVM-based estimation of branching scores.	Proves that expensive solver heuristics (like SB) can be replaced by fast ML approximations.
Bengio et al. (2021)	A methodological roadmap for ML in CO, distinguishing between learning by demonstration and experience .	Reinforcement Learning (RL) vs. Imitation Learning; End-to-end vs. hybrid architectures.	Foundational methodology; ensures our hybrid solver approach follows principled ML design.

Week- 1,2

PAPERS READ:

Paper / Source	Core Idea / Contribution	Key Techniques	Relevance to Dissertation
Mandi & Guns (2020)	End-to-end Predict+Optimize: Differentiates through LP layers using a principled logarithmic barrier term.	Interior Point Method; Homogeneous self-dual formulation.	Offers a way to train the solver based on "regret" (how much a prediction hurts the final objective).
ML4CO-Bench-101 (2025)	Unified benchmark and taxonomy for neural CO solvers (Global Prediction, Local Construction, Adaptive Expansion).	One-Shot (OS) vs. Generative (Gen) models; comprehensive dataset suites.	Provides standardized metrics and comparison baselines for benchmarking crossword solver speed.

Week- 3

PAPERS READ:

Paper / Source	Core Idea / Contribution	Key Techniques	Relevance to Dissertation
BackPaS (2026)	Redefines prediction to identify backbones (variables fixed across all optimal solutions) rather than single solutions.	8-layer Bipartite Graph Transformers; Adaptive Trust Regions based on confidence.	Theoretical justification for "Certainty-First" branching; proves structural patterns generalize to larger puzzles.
O'Connor et al.	Learning to Prune (LTP) : Uses a classifier to fix or prune vertices, significantly simplifying the instance.	Multiplicative Weights Update (MWUA) as a fast surrogate for LP features.	Provides a "fast feature" engine (MWUA) to scale your solver to massive crossword grids.
Zhang et al. Survey (2023)	Comprehensive survey categorizing ML for MIP into exact (branch-and-cut) and approximate (heuristics) solving.	Taxonomy of Graph Neural Networks (GNNs) for Bipartite/Tripartite graph encoding.	Broad academic context; helps classify our solver within the wider ML4CO research field.

Week- 4

PAPERS READ:

Paper / Source	Core Idea / Contribution	Key Techniques	Relevance to Dissertation

Ongoing: Current Week
Next Update on: 12-6-26